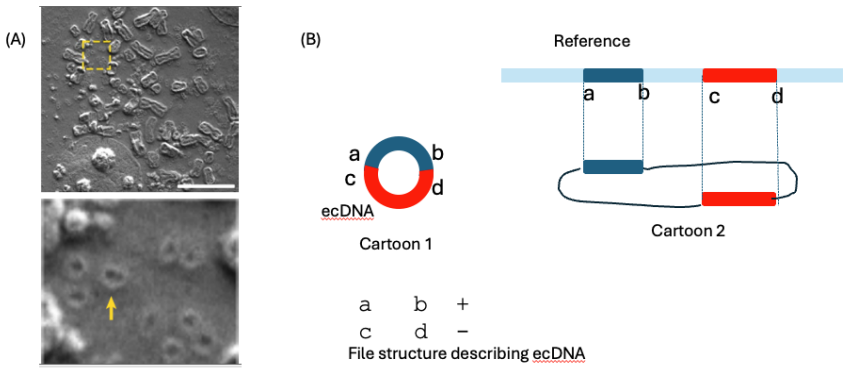
# CSE283/BENG203: Assignment 2

## Logistics

Submit a zip archive of your code and a README file by Gradescope. The README should explain exactly how to run your scripts.

## RNA-seq and database filtering




1. **RNA-seq analysis**. Extra-chromosomal DNA are circular DNA structures often seen in cancer samples. Panel (A) describes an electron-microscope image in a metaphase cell where the chromosomes are compacted and the circular structures can be easily seen. Panel (B) describes the genomic elucidation of an ecDNA, where two segments from a chromosome are joined to form the ecDNA. Using reference coordinates, the ecDNA can be described in a 'bed'-format file structure, whch provides the begin and end coordinates of each segment and their orientation (See panel B). The ecDNA can also be visually represented using cartoons 1 and 2. You are given an ecDNA structure as a bed-formatted file in this assignment.

   EcDNA have highly accessible chromatin resulting in *dysregulation* of gene expression, and we will study that in this assignment. In addition to the structure of an ecDNA in a bed-file format, you are given the location of some but not all genes on the ecDNA, also in a bed-file format; and gene expression data as the count of the number of mRNA sequences in non-overlapping 50bp bins for the genome. All coordinates are on the hg38 assembly. For better understanding, you can also visualize the ecDNA region on the UCSC genome browser.

   (a) Plot a cartoon (using cartoon 2 format) of the ecDNA structure using the given bed file. The x-axis should depict reference genomic coordinates, and the ecDNA should be depicted as intervals connected by lines.

   (b) Identify the protein-coding gene on the ecDNA with the highest expression. To answer this question, you may need to download the exon coordinates of the ecDNA genes from the genome browser. You can use the table-browser on the UCSC genome browser site for this purpose. Provide some reasons for why this gene is the most expressed gene.

   (c) Using the gene boundaries for LPIN1 in the provided bed-file, and its exon coordinates, plot its gene expression values in 50bp bins along the length of the transcript. Note that you are given the expression data on genomic coordinates, but here, we are asking you to map them to the transcript. Compute the mean of the expression values per bin, $\lambda$. Assuming that the counts in each bin are Poisson distributed with parameter $\lambda$, report the number of bins that have significantly higher or lower counts. **No multiple testing correction is needed. Just compute the P-value for each bin, and show it on the plot**.

   (d) Is there expression of a *non-genic* region? In order to answer this question, you may need to obtain the coordinates of all genes (coding/non-coding) on the ecDNA region.

(e) The previous two questions provide examples of dysregulation of genes on ecDNA. Suggest how the structural variation in ecDNA could explain the dysregulation (**full credit will be given for all reasonable answers.**)

2. *Database filtering.* Many sequence mapping tools use a 'seed and extend' filtering strategy similar to the one discussed in class. Describe the differences between the following tools: Blast, Blat, bowtie (or, bowtie2), minimap2. For example, you could describe the following, but also any additional strategy that may have been used.

   (a) The indexing method used for fast filtering.
   (b) Are the queries indexed or the database?
   (c) keyword size for seeding
   (d) algorithmic improvements for alignment
   (e) Target for the alignment. What is the level of similarity that the method aims to retrieve from the database.

   Explain how setting a target for the alignment (e.g. Find all homologs in a related species, or find the mapping of human RNA to the human genome) has an influence on the design of the parameters and the final algorithm.

3. **Bloom filters.** Please use a word-processor (preferably latex) with an equation editor to write your answers, or write them cleanly, so it is easier to understand. As discussed in class, we use a bit string $B$ of $m$ bits (initially all 0s) to store a collection of $n$ objects, by setting $h$ bits to 1 for each object.

   (a) Given a query object $o$, derive an expression for the probability $P_{\text{FP}}$ that the Bloom Filter returns an incorrect positive answer to the membership query.
   (b) Given $m, n$ derive an expression for $h$ s.t. $P_{\text{FP}}$ is minimized.
   (c) Using the value of $h$ derived in the previous problem, derive a lower bound on $m$ to achieve the desired false membership rate $P_{\text{FP}}$. Next, consider a database of $10,000$ reads, each of length 99. Your goal is to make a Bloom Filter that stores all 25-mers in all reads. Make a table that records the minimum size of the BF in kb (kilo-bytes), for different error probabilities $P_{\text{FP}}$, when querying with keywords. Use $P_{\text{FP}} \in \{0.001, 0.01, 0.05, 0.1, 0.5\}$.
   (d) Create a Bloom Filter with the small 10,000 read database where each object is a keyword of length 25. Describe the design of the filter (selection of $m, h$), and how you did the hashing (In Python you can try mmh3 with h different seeds selected once). The database is human, and the query transcripts were sampled from a human cervical cancer cell-line. Identify the human query sequences. For the non-human queries, what are they, and do they reveal any useful information?